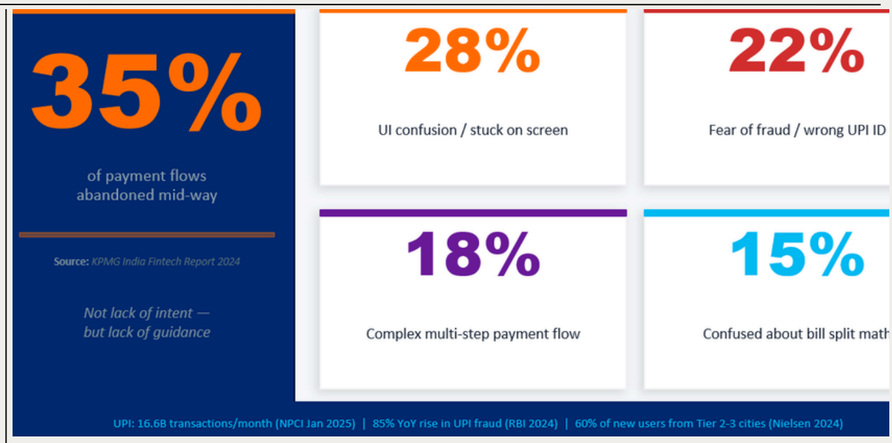
Paytm Copilot

FIN-O-HACK
fintech hackathon

Problem Track - AI for Paytm Users

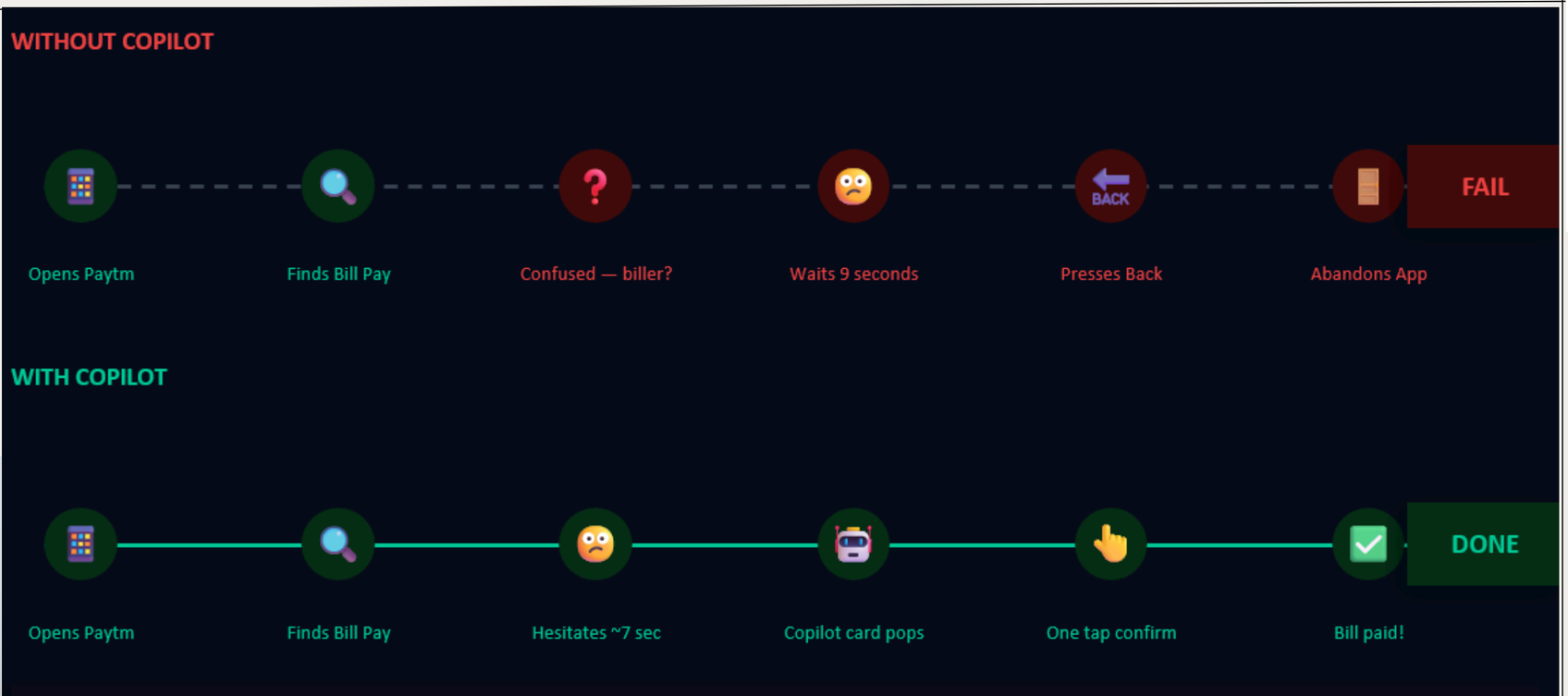
Problem - 35% of payment flows are abandoned mid-way — because Paytm watches but never helps.

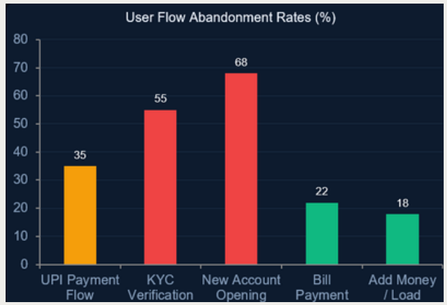
Solution - We gave Paytm a brain — one that watches, learns, and acts so users never get stuck again.



60% of semi-literate users struggle with multi-step UPI flows

0 UPI apps today offer proactive AI help when users are stuck





Team - PayThis
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Solution: What is Paytm Copilot:

An invisible AI layer inside Paytm that watches what you're doing, understands your intent, and proactively offers to complete the task — in one tap

How Copilot Triggers (Real Scenario)

1. User opens **Send Money**
→
2. Enters contact then hesitates
→
3. 7 seconds pass (confusion detected)
→
4. **Copilot appears** Task done instantly

- GPS doesn't wait for you to ask "which turn?" It watches and speaks before you need it. Copilot is GPS — for every screen in Paytm.

3 Live Demo Scenarios

01 Bill Split Confusion

User stuck → AI detects confusion → suggests bill split instantly.

02 Unknown UPI Fraud

AI detects risky payment → warns user before transaction.

03 Voice Command Flow

"Pay Rahul 500" → AI fills details → ready to confirm.

Watches, Understands, Acts — before you even ask.

✗ Normal Chatbot

- User must manually open it
- User types a question
- Waits for a text response
- Reads the reply carefully
- Navigates manually to act
- Task done — maybe
- Reactive, not intelligent

VS

✓ Paytm Copilot

- Nobody opens it — watches passively
- Detects confusion from behaviour
- Appears precisely when useful
- Shows one-tap action card
- Completes the task instantly
- Always done — guaranteed
- Proactive ambient intelligence

Features: What Copilot Does

The Features — Built for 300M Users

Behavioural AI

Confusion Detection

Trigger: Stuck 7+ sec on screen

Reads screen state + dwell time → classifies intent → surfaces a relevant nudge.

Automation

One-Tap Completion

Trigger: Partial action detected

Fills form, selects contacts, sets amount — user just confirms.

NLU

Voice / Text Commands

Trigger: User types in search bar

"Pay Rahul 500 for lunch" → Copilot resolves contact + prefills form. No navigation.

Personalisation

Smart Reminders

Trigger: Recurring payment pattern

"You pay BSNL every 5th — want me to pay it tomorrow?" Learns habits silently.

Trust & Safety

Fraud Micro-Check

Trigger: Unusual transaction detected

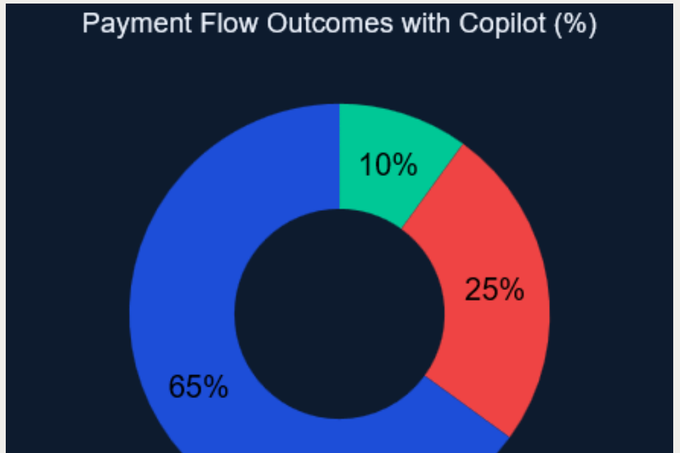
"You've never paid this UPI ID before and it's over ₹5,000. Confirm this is intentional?"

Inclusion

Multilingual Support

Trigger: Device language preference

All Copilot suggestions in Hindi, Tamil, Telugu, Marathi & more.



300M — 300M MORE PAYMENTS

users × 1 saved txn/month =

10% — MILLIONS MORE TXNS/DAY

abandonment recovery →

↑DAU — HIGHER DAILY ACTIVE USAGE

habit-forming nudges →

Fraud Shield

Trust scores, alerts, and fraud detection

Fraud Shield

Trust scores, alerts, and fraud detection

Bill Splitting

Quick splits & auto suggestions

Group Messages

Auto notify & remind friends

Analytics

Weekly & monthly spending insights

Voice Payments

Pay hands-free in 15 languages

Feature	Paytm Copilot	Phone Pe	GPay	BHIM
Confusion Detection	✅	❌	❌	❌
Voice / NLU	✅	❌	Partial	❌
Smart Reminders	✅	❌	❌	❌
Fraud Micro-Nudge	✅	Basic	Basic	❌
Multilingual AI	✅	❌	❌	❌

Architecture: How It Works

Privacy first · On-device intelligence · Cloud LLM only when confidence > 0.7

1 — Event Listener SDK
React Native · Flutter
Screen ID
Dwell time
Last 3 actions
Form fill state

→

2 — On-Device Classifier
TF Lite / ONNX · 2MB
Screen+dwell+actions
→ Intent label
Runs fully OFFLINE

→

3 — Threshold Gate
Confidence > 0.7 only
Below threshold:
Do nothing silently
Avoids false positives

→

4 — LLM Suggestion API
FastAPI · GPT-4o mini / Haiku
Intent+context
→ Suggestion text
+ Action payload

→

5 — Overlay UI + Execute
Non-blocking card
User taps YES:
Form filled & done
Dismiss = logged

Decision Flow — How Copilot Fires

Entire decision tree executes in <800ms · On-device until confidence confirmed · No cloud call below threshold

Privacy Architecture — Zero PII to Cloud

What stays on device vs what reaches cloud

🔒 ON-DEVICE (stays here)

- Raw screen state & UI interactions
- Dwell time measurements
- TF Lite intent classifier (2MB)
- Confidence score computation
- Partial form field states
- User payment behavior patterns
- Language preference (device)

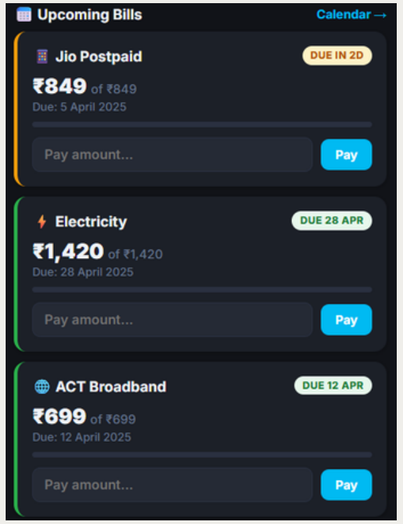
IF > 0.7

☁️ CLOUD (anonymised only)

- ✓ Intent label (e.g. "bill_split")
- ✓ Screen context ID (no values)
- ✓ Language preference code
- ✓ Past accepted suggestions
- ❌ No UPI IDs or phone numbers
- ❌ No transaction amounts
- ❌ No contact details

Prototype Explanation

- **Bill Reminders**



Smart Due-Date Tracking

Bills auto-fetched & sorted by due date — Jio (2 days), ACT Broadband (Apr 12), Electricity (Apr 28).

Urgency Color Codes

Orange border = due soon. Green border = upcoming. Lets users prioritise at a glance instantly.

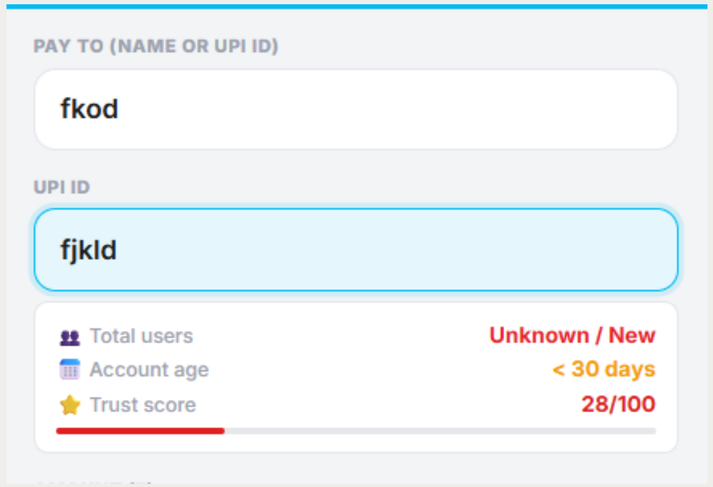
One-Tap Direct Payment

Each card shows exact amount owed with a pre-filled 'Pay' button — zero navigation required.

Partial Payment Support

User can edit the pay amount field to make partial payments if full amount isn't available.

- **Unknown & Fraudulent UPI Detection**



Live Trust Score Check

Every UPI ID is scored 0–100 at transaction time. Score 28/100 triggers a red warning banner.

Account Age Verification

Accounts under 30 days are highlighted as suspicious — a common trait of scam / mule accounts.

Unknown Recipient Alert

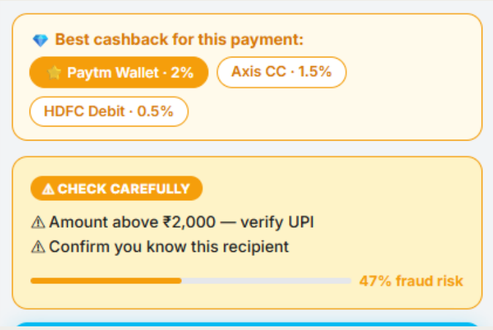
If the UPI ID has never transacted before, it's flagged 'Unknown/New' in bright red immediately.

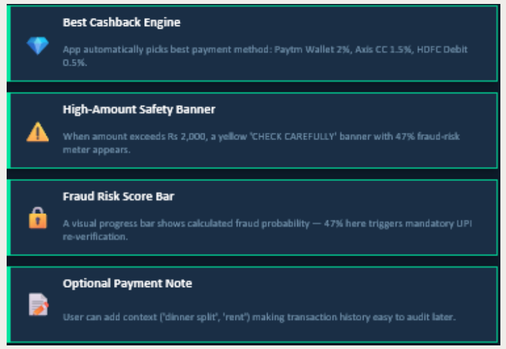
Double-Check Prompt

User is asked to confirm they know the recipient before proceeding — preventing accidental fraud.

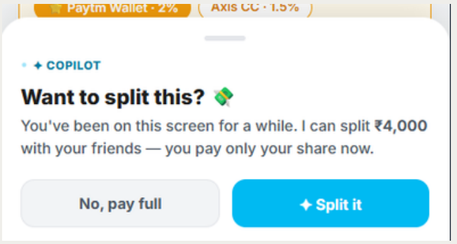
Prototype Explanation

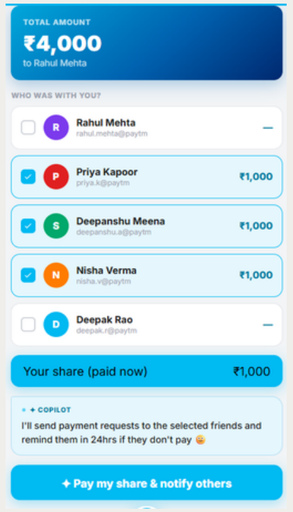
- **Cashback Offers & High - Amount Safety Check**

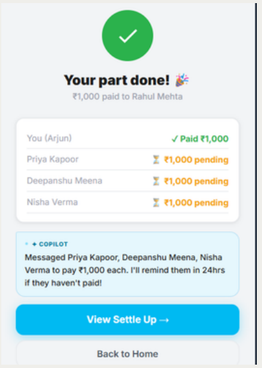


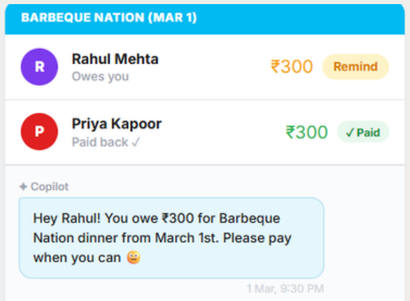


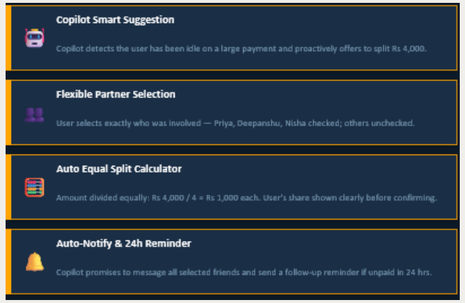
- **Bill Splitting**

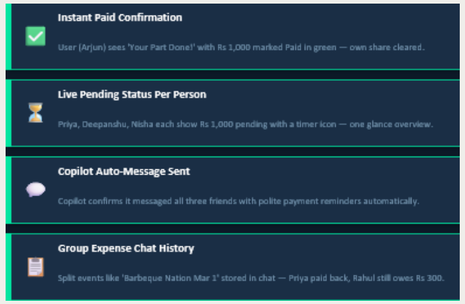






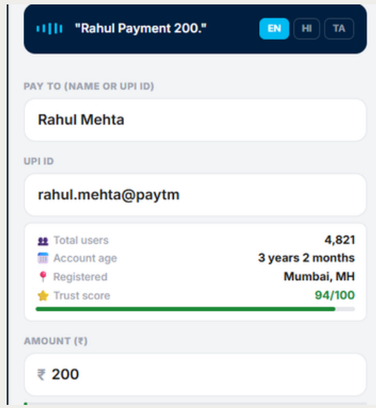




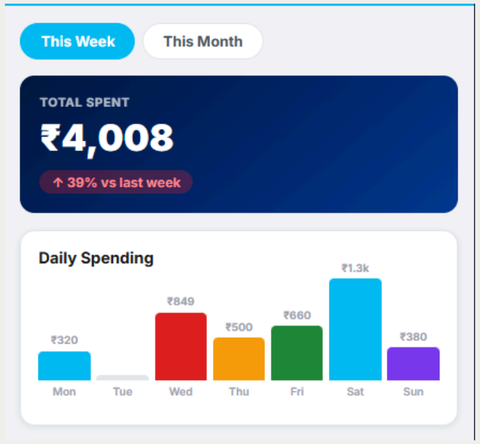


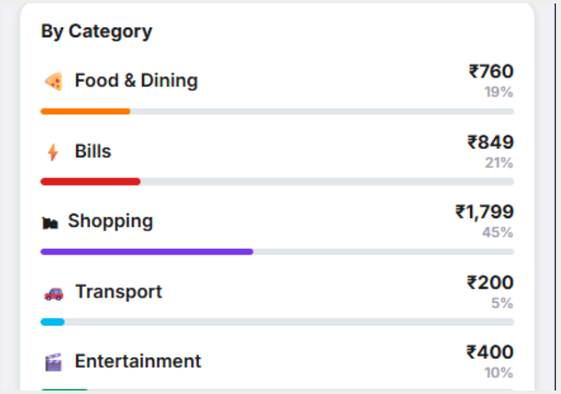
Prototype Explanation

- **Voice - Powered Payment**



- **Spending Analysis Dashboard**





Impact + Why Paytm Wins

So what? Why does this matter?

ON-DEVICE (stays here)

- Raw screen state & UI interactions
- Dwell time measurements
- TF Lite intent classifier (2MB model)
- Confidence score computation
- Partial form field states
- User's payment history patterns
- Language preference (OS locale)

IF >0.7

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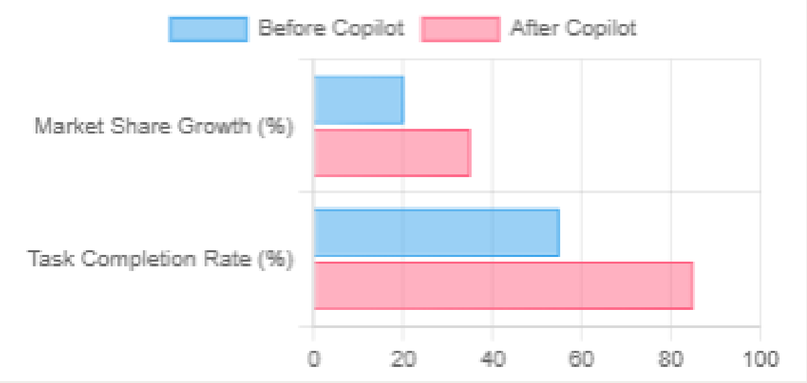


The best UX knows what you need before you do.""Paytm Copilot makes that real — for 300M users, starting today.

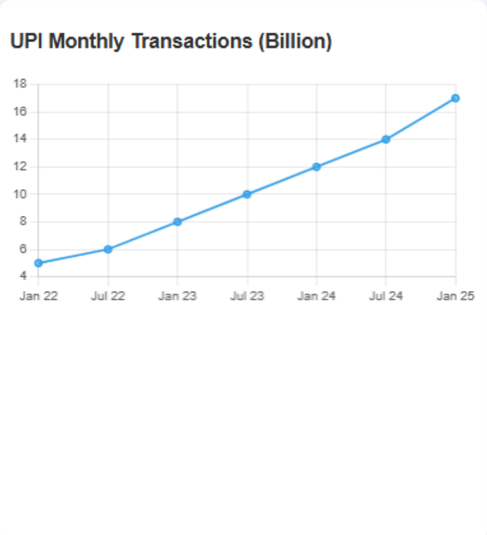
Why Now? Why Paytm?

- LLMs are finally fast + cheap enough for real-time in-app use
- 300M Paytm users = unmatched training signal for intent detection
- Zero competitors have shipped ambient AI in UPI — first-mover wins
- On-device ML makes privacy story compelling for regulators & users

Avg Paytm txn value ₹1,850	*Industry average*
Copilot API cost / suggestion ~₹0.018	*Claude Haiku pricing*
Suggestions per recovered txn ~3	*Based on demo flow*
Cost to recover one txn ₹0.054	*< 1 paisa effectively*
Revenue per recovered txn ₹9–18	*0.5–1% MDR on ₹1,850*

The UPI Market Opportunity

India's UPI is growing rapidly — massive scale for AI impact



₹200T+ UPI GMV annually

16.6B Monthly transactions

500M+ Smartphone users

#1 India leads UPI globally

If AI recovers just 1% of failed payments → 160M extra successful transactions/month

The Flywheel Effect

1. User interacts with Copilot
2. Accept / dismiss logged locally
3. Classifier retrained per cluster
4. Next suggestion is more accurate

Week 12: 84% acceptance rate · 3% false positives · Users stop abandoning — they wait for Copilot

The Math

300M users

× 1 recovered txn / month
= 300M more payments

Avg txn ₹1,850 × 0.5% MDR

= ₹2,775 Cr / year

LLM cost: ₹0.054 / recovery

ROI(Return of Investment) = 166–333x

References

#	Source	Data Used	URL / Reference
1	**NPCI (National Payments Corporation of India)**	UPI monthly transaction volume (16.6B txns, Jan 2025) · UPI MDR rate (0.5% P2M) · UPI annual GMV (■200T+)	*npci.org.in → Statistics → UPI Product Statistics (public, updated monthly)*
2	**RBI Annual Report 2023-24**	UPI fraud cases rose 85% YoY · Average UPI transaction value ■1,850 · India payment system overview	*rbi.org.in → Publications → Annual Report 2023-24 · Chapter on Payment Systems*
3	**KPMG India Fintech Report 2024**	35% payment flow abandonment rate · Root causes of abandonment · India fintech investment trends	*kpmg.com/in → Insights → KPMG India Fintech Report 2024 (available as free PDF download)*
4	**Paytm Q3 FY2024 Earnings Report**	300M+ registered users · Monthly transacting users (MTU) · Wallet balance data	*paytm.com/investor-relations → Quarterly Results → Q3 FY2024 Earnings Presentation*
5	**Baymard Institute (2024)**	28% of abandonment due to UI confusion · Checkout abandonment benchmarks · UX research data	*baymard.com → Research → Checkout Usability (Industry benchmark, referenced by 4,000+ companies)*
6	**Nielsen India Mobile Payment Behaviour Study 2024**	60% of new UPI users are from Tier 2-3 cities · Digital literacy gap data · Language preference in payments	*nielsen.com → India Insights → Mobile Payment Study 2024*
7	**Groq Inc. (API documentation)**	Free tier: 14,400 requests/day · Llama 3 8B latency: ~200-300ms · Pricing: $0.05 per million tokens	*console.groq.com → Pricing & Docs (free registration required)*
8	**Google (Gmail Smart Reply blog)**	Smart Reply used in 25% of mobile email replies · Ambient AI effectiveness data	*blog.google → 'Gmail Smart Reply is being used for 25% of replies in Gmail mobile app' (2019)*
9	**RBI Payment Vision 2025 Document**	Target: 40% reduction in failed digital payments · AI-based fraud prevention push	*rbi.org.in → Publications → Payments Vision 2025*
10	**TensorFlow Lite Documentation**	On-device model size benchmarks (~2MB for classification) · Android inference latency (<50ms on mid-range)	*tensorflow.org/lite → Performance → Benchmarks*

Website Link :- Paytm Copilot

Github Repository Link :- Github Repository

Video Link :- Video Explanation